



✦ Zepto SQL Data Analyst project

Tools: PostgreSQL, pgAdmin, SQL

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Project introduction

About the project

- This project analyzes a real-world inventory dataset using PostgreSQL.
- The goal is to clean the data, explore inventory information, and generate business insights using SQL..
- The project simulates real-world tasks performed by Data Analyst.

Problem statement

business problem

This project aims to solve the following business challenges:

- Understand inventory distribution.
- Analyze product pricing and discounts.
- Identify out-of-stock products.
- Evaluate category-wise performance.
- Generate insights to support business decision.

```
--data analysis
```

```
-- Q1. Find the top 10 best-value products based on the discount percentage.
```

```
SELECT DISTINCT name, mrp, discountPercent
```

```
FROM zepto
```

```
ORDER BY discountPercent DESC
```

```
LIMIT 10;
```

	name character varying (150)	mrp numeric (8,2)	discountpercent numeric (5,2)
1	Dukes Waffy Orange Wafers	4500.00	51.00
2	Dukes Waffy Strawberry Wafers	4500.00	51.00
3	Dukes Waffy Chocolate Wafers	4500.00	51.00
4	Epigamia Fruit Yogurt Alphonso Mango	4000.00	50.00
5	Epigamia Fruit Yogurt Vanilla	4000.00	50.00
6	Chef's Basket Durum Wheat Elbow Pasta	16000.00	50.00
7	Chef's Basket Durum Wheat Penne Pasta	16000.00	50.00

```
--Q2.What are the Products with High MRP but Out of Stock
SELECT DISTINCT name, mrp
FROM zepto
WHERE outOfStock = TRUE and mrp > 300
ORDER BY mrp DESC;
```

	name character varying (150)	mrp numeric (8,2)
1	Patanjali Cow's Ghee	56500.00
2	MamyPoko Pants Standard Diapers, Extra Large (12 - 17 kg)	39900.00
3	Aashirvaad Atta With Mutigrains	31500.00
4	Everest Kashmiri Lal Chilli Powder	31000.00
5	Hershey's Cocoa + Almond Spread	29500.00
6	Madhur Pure And Hygienic Sugar	29500.00
7	RRO Mozzarella Block Cheese	29500.00

--Q3.Calculate Estimated Revenue for each category

```
SELECT category,  
SUM(discountedSellingPrice * availableQuantity) AS total_revenue  
FROM zepto  
GROUP BY category  
ORDER BY total_revenue:
```

	category character varying (120) 🔒	total_revenue numeric 🔒
1	Fruits & Vegetables	1084600.00
2	Meats, Fish & Eggs	2069300.00
3	Biscuits	2500760.00
4	Beverages	5505100.00
5	Dairy, Bread & Batter	5505100.00
6	Health & Hygiene	6418000.00
7	Home & Cleaning	12266100.00

-- Q4. Find all products where MRP is greater than ₹500 and discount is less than 10%.

```
SELECT DISTINCT name, mrp, discountPercent
FROM zepto
WHERE mrp > 500 AND discountPercent < 10
ORDER BY mrp DESC, discountPercent DESC;
```

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	name character varying (150)	mrp numeric (8,2)	discountpercent numeric (5,2)
1	Dhara Kachi Ghani Mustard Oil Jar	125000.00	8.00
2	Saffola Gold (Jar)	124000.00	0.00
3	Fortune Rice Bran Health Oil (Jar)	105000.00	1.00
4	Dhara Filtered Groundnut Oil (Jar)	105000.00	1.00
5	Dhara Filtered Groundnut Oil (Jar)	105000.00	0.00
6	Fortune Soyabean Oil	100500.00	0.00
7	Fortune Sunlite Refined Sunflower (Jar)	92500.00	0.00

```
-- Q5. Identify the top 5 categories offering the highest average discount percentage.  
SELECT category,  
ROUND(AVG(discountPercent),2) AS avg_discount  
FROM zepto  
GROUP BY category  
ORDER BY avg_discount DESC  
LIMIT 5;
```

	category character varying (120) 🔒	avg_discount numeric 🔒
1	Fruits & Vegetables	15.46
2	Meats, Fish & Eggs	11.03
3	Ice Cream & Desserts	8.32
4	Chocolates & Candies	8.32
5	Packaged Food	8.32

```
-- Q6. Find the price per gram for products above 100g and sort by best value.
SELECT DISTINCT name, weightInGms, discountedSellingPrice,
ROUND(discountedSellingPrice/weightInGms,2) AS price_per_gram
FROM zepto
WHERE weightInGms >= 100
ORDER BY price_per_gram;
```

	name character varying (150)	weightingms integer	discountedsellingprice numeric (8,2)	price_per_gram numeric
1	Vicks Cough Drops Menthol	1160	2000.00	1.72
2	Aashirvaad Iodised Salt	1000	1900.00	1.90
3	Onion	3000	5700.00	1.90
4	Onion	1000	2100.00	2.10
5	Tata Salt	1000	2400.00	2.40
6	Shubh kart - Nirmal sugandhi mogra wet dhoop zipper 20 sticks	1160	2800.00	2.41
7	Beetroot	500	1300.00	2.60


```
--Q7.Group the products into categories like Low, Medium, Bulk.
SELECT DISTINCT name, weightInGms,
CASE WHEN weightInGms < 1000 THEN 'Low'
      WHEN weightInGms < 5000 THEN 'Medium'
      ELSE 'Bulk'
      END AS weight_category
FROM zepto;
```

	name character varying (150)	weightingms integer	weight_category text
1	Popular Essentials Poppy Seeds (Gasagase)	100	Low
2	Origami Kitchen Towels 2 In 1 Quilted 2ply	116	Low
3	Origami Good Karma Paper Serviettes - 100 napkins (1 ply)	58	Low
4	Johnson's Baby Top To Toe Bath Body Wash	210	Low
5	Harpic Hygienic Toilet Cleaner Rim Block Citrus	26	Low
6	Lizol Disinfectant Surface & Floor Cleaner Liquid Jasmine	500	Low
7	Sapota	348	Low

--Q8.What is the Total Inventory Weight Per Category

```
SELECT category,  
SUM(weightInGms * availableQuantity) AS total_weight  
FROM zepto  
GROUP BY category  
ORDER BY total_weight;
```

	category character varying (120) 	total_weight bigint 
1	Meats, Fish & Eggs	48016
2	Biscuits	84431
3	Fruits & Vegetables	91794
4	Health & Hygiene	142904
5	Dairy, Bread & Batter	143735
6	Beverages	143735
7	Paan Corner	348187

Key Insights

Highest discount categories.

Products with high MRP and low discounts.

Out-of-stock products.

Revenue contribution by category.

Best value products based on price per gram

Conclusion

Successfully built and managed a PostgreSQL database.
Cleaned and analyzed a real-world e-commerce dataset.

Solved business problems using SQL queries.

Gained practical experience in data analysis and SQL.